

4.11.31

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Problem 1 (4.11.31). Find the area of the region bounded by the line $y = 3x + 2$, the X axis and the ordinates $x = -2$ and $x = 1$.

Solution: The total area is the sum of the areas of two triangles. To find their vertices, we must find the intersection points of the boundary lines: $y = 3x + 2$ (the line), $y = 0$ (the axis), $x = -2$, and $x = 1$.

Finding the Vertices via Gaussian Elimination

1. Intersection with the x-axis (Vertex B):

This is the intersection of $3x - y = -2$ and $y = 0$. The augmented matrix is:

$$\left(\begin{array}{cc|c} 3 & -1 & -2 \\ 0 & 1 & 0 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + R_2} \left(\begin{array}{cc|c} 3 & 0 & -2 \\ 0 & 1 & 0 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1/3} \left(\begin{array}{cc|c} 1 & 0 & -2/3 \\ 0 & 1 & 0 \end{array} \right) \Rightarrow \mathbf{B} = \begin{pmatrix} -2/3 \\ 0 \end{pmatrix} \quad (0.1)$$

2. Intersection with the ordinate $x = -2$ (Vertex D):

This is the intersection of $3x - y = -2$ and $x = -2$. The augmented matrix is:

$$\left(\begin{array}{cc|c} 3 & -1 & -2 \\ 1 & 0 & -2 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 3 & -1 & -2 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & -1 & 4 \end{array} \right) \xrightarrow{R_2 \rightarrow -R_2} \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -4 \end{array} \right) \Rightarrow \mathbf{D} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} \quad (0.2)$$

3. Intersection with the ordinate $x = 1$ (Vertex E):

This is the intersection of $3x - y = -2$ and $x = 1$. The augmented matrix is:

$$\left(\begin{array}{cc|c} 3 & -1 & -2 \\ 1 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 3 & -1 & -2 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & -1 & -5 \end{array} \right) \xrightarrow{R_2 \rightarrow -R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 5 \end{array} \right) \Rightarrow \mathbf{E} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \quad (0.3)$$

The remaining vertices are trivial intersections with the axes: $\mathbf{A} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. These points form two triangles: $\triangle ABD$ and $\triangle EBC$.

Area Calculation

The total area is the sum of the areas of $\text{ar}(\triangle ABD)$ and $\text{ar}(\triangle EBC)$.

$$\text{ar}(\triangle ABD) = \frac{1}{2} \| (\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{D}) \| = \frac{1}{2} \left\| \begin{pmatrix} -4/3 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 4 \end{pmatrix} \right\| = \frac{8}{3} \quad (0.4)$$

$$\text{ar}(\triangle EBC) = \frac{1}{2} \| (\mathbf{E} - \mathbf{C}) \times (\mathbf{B} - \mathbf{C}) \| = \frac{1}{2} \left\| \begin{pmatrix} 0 \\ 5 \end{pmatrix} \times \begin{pmatrix} -5/3 \\ 0 \end{pmatrix} \right\| = \frac{25}{6} \quad (0.5)$$

The total area of the region is the sum of the two triangle areas.

$$\text{Area of the region} = \frac{8}{3} + \frac{25}{6} = \frac{41}{6} \quad (0.6)$$

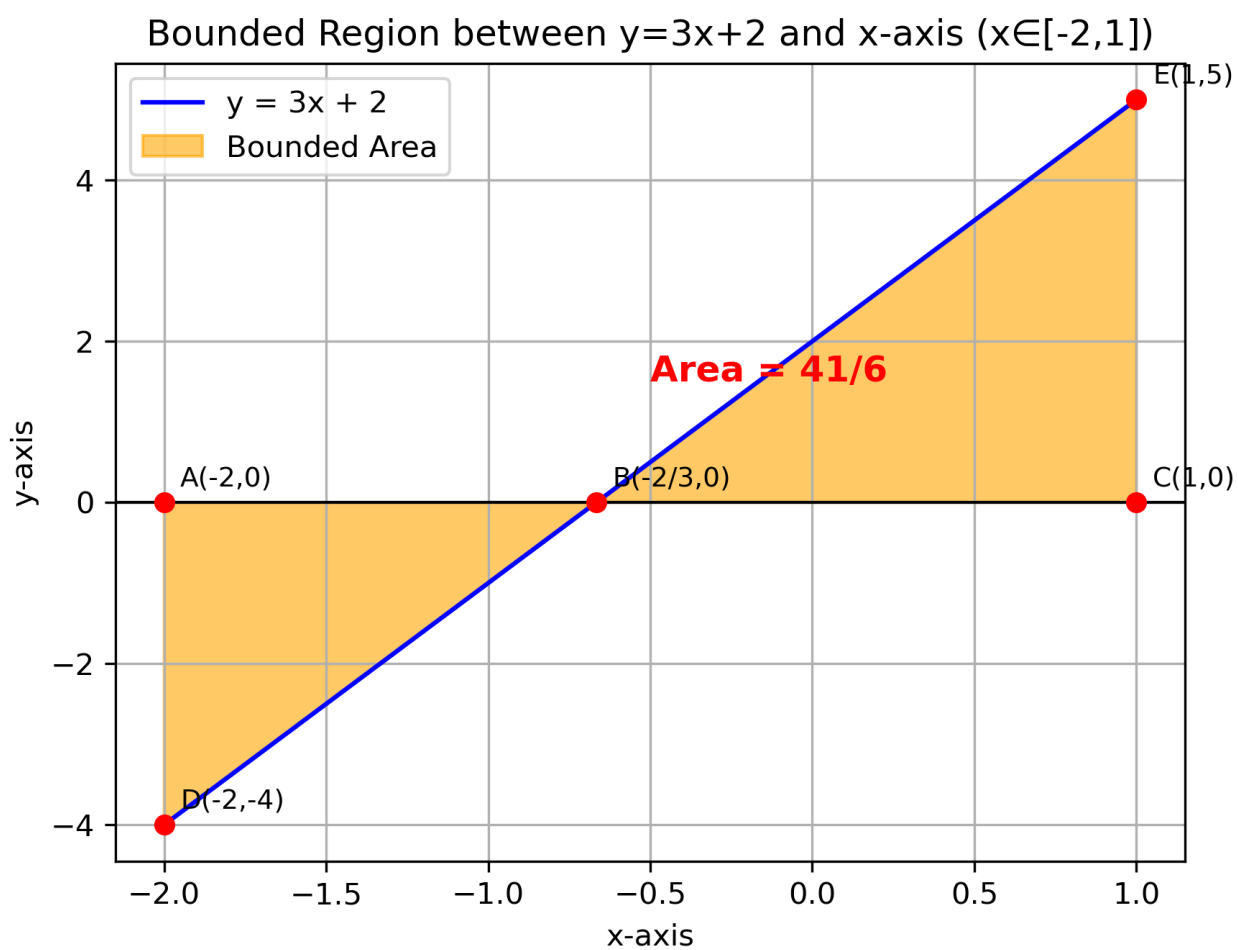


Fig. 0.1: Region bounded by the line and ordinates